Chapter 2

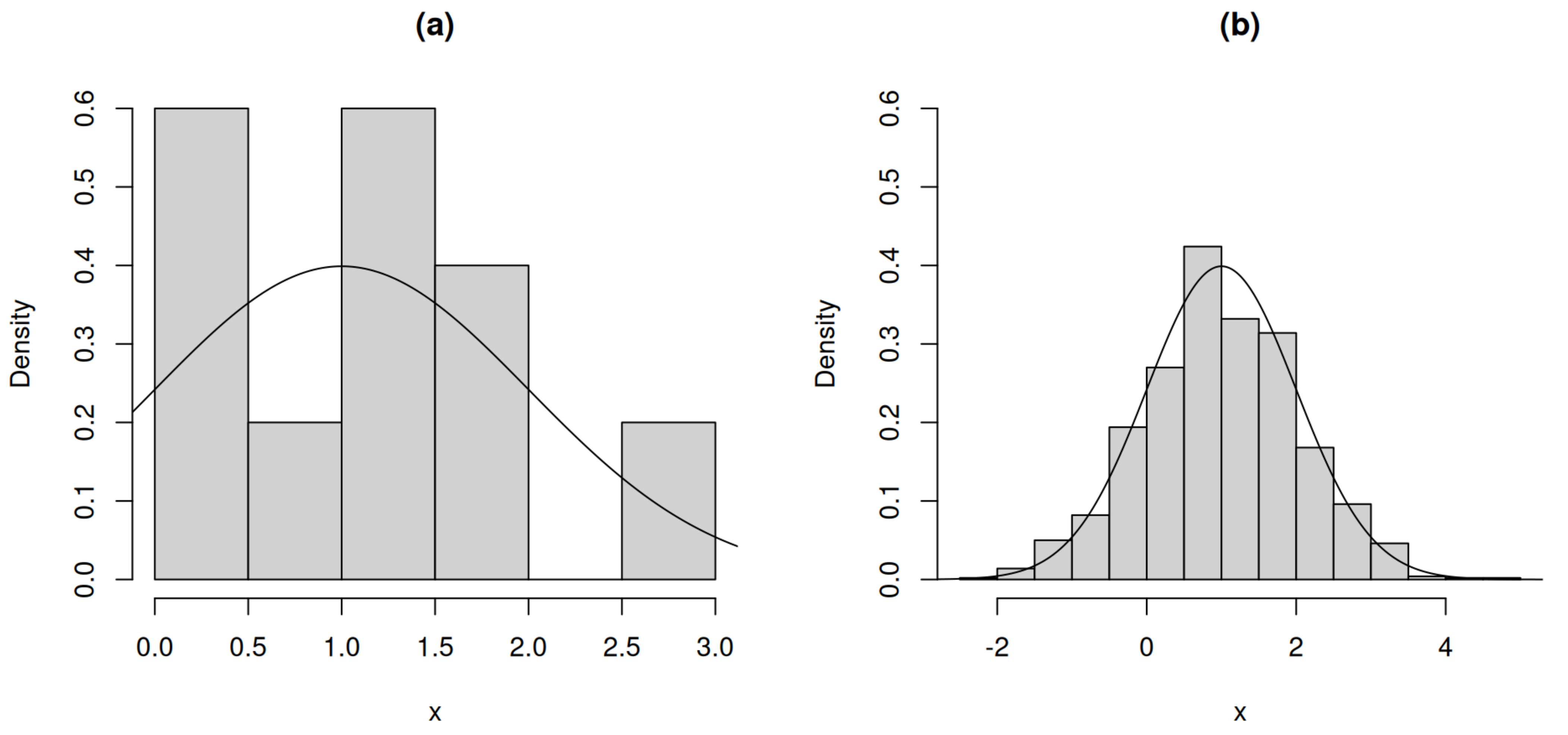
Qualities of Estimators: Defining Good, Better, and Best

Estimation

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, 1)$.

If n is large, then a histogram of the sampled values should look approximately like the distribution from which the sample came.

Estimation example



(a) is a histogram of a random sample of 10 values from a N(1, 1) distribution. (b) is a histogram of a random sample of 1,000 values from a N(1, 1) distribution

Estimation approach

A natural approach for estimating the mean is to use the **sample mean**

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

- The notation $\hat{\mu}$ denotes an estimator of the parameter μ.
- We write $\hat{\mu} = \bar{X}$ to indicate that $\bar{X}$ is the estimator of μ.

Estimator vs estimate

An **estimator** is a random variable.

- It is a formula that tells us what to do when we get the data in the future.
- Uppercase $\bar{X}$ indicates that the estimator is a random variable.

An **estimate** is a single number obtained from the observed sample.

- It is not random since it is a function of the observed data.
- Lowercase $\bar{x}$ indicates that the estimate is not random.

Estimation ponderings

Estimating the mean μ with the sample mean $\bar{X}$ seems sensible, but leaves some questions unaddressed.

1. How “good” is the estimator?
2. What does it mean for an estimator to be “good”?
3. Can we find a better estimator?
4. How do we choose an estimator in other contexts, like the α parameter from a $\mathrm{Gamma}(\alpha, \beta)$ distribution?

2.1
Notation, Statistics, and Unbiasedness

Estimation context

Let θ denote a parameter or parameter vector.

- E.g., $\theta = \lambda$ for an exponential distribution.
- E.g., $\theta = (\alpha, \beta)$ for a Gamma distribution.

Estimation context

Distributions depends on parameters and we will emphasize that by explicitly including the relevant parameters in the description of the cdf, pmf, or pdf.

- E.g., The pdf might be denoted $f(x;\theta)$ or $f(x \mid \theta)$.
- E.g., If $X_1, X_2 \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$, then the joint pdf is

$$f(x_1, x_2; \mu, \sigma^2) = f(\vec{x}; \mu, \sigma^2).$$

Estimation goal

Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim} F(x;\theta)$.

Our goal is to estimate a parameter θ or a function of θ, $\tau(\theta)$, using a **statistic**.

A **statistic** is a function that only depends on the data and known values.

What is a statistic?

A statistic will be denoted by $T = t(X_1, X_2, \ldots, X_n) = t(\vec{X})$.

- $T = t(\vec{X}) = \bar{X}$ is a one-dimensional statistic.
- $T = t(\vec{X}) = (\bar{X}, \sum_{i=1}^{n} X_i^2, X_{(1)})$ is a multi-dimensional statistic.

Definition 2.1.1 (Unbiased)

An estimator T is **unbiased** for $\tau(\theta)$ if

$$E[T] = \tau(\theta).$$

Mean of sample mean

Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim} F(x; \theta)$, with $E(X) = \mu < \infty$.

Prove that

$$E(\bar{X}) = \mu.$$

$\bar{X}$ is an unbiased estimator of μ.

Proof

Variance of sample mean

Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim} F(x;\theta)$, with $\operatorname{var}(X) = \sigma^2 < \infty$.

Prove that

$$\operatorname{var}(\bar{X}) = \sigma^2/n.$$

Proof

2.2

Mean Squared Error and Bias

Definition 2.2.1 (Bias)

The **bias** of an estimator $\hat{\theta}$ of θ is

$$B(\hat{\theta}) = E(\hat{\theta}) - \theta.$$

Definition 2.2.2 (Mean Squared Error)

The mean squared error (MSE) of an estimator $\hat{\theta}$ of θ is

$$\mathrm{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2].$$

Bias of an unbiased estimator

The bias of an unbiased estimator $\hat{\theta}$ of θ is zero.

Proof

MSE of an unbiased estimator

If $\hat{\theta}$ is an unbiased estimator of θ, then

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{var}(\hat{\theta}).$$

Proof

MSE, variance, and bias relationship

If $\hat{\theta}$ is an estimator of θ, then

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{var}(\hat{\theta}) + [B(\hat{\theta})]^2.$$

Proof

Connecting the dots

- If multiple estimators are unbiased, we prefer the estimator with the smaller variance since it tends to be closer to to the truth, on average.
- If we are comparing two estimators, one biased and one unbiased, then the MSE is a fairer measure of each estimators quality.
- In the next section, we will consider other ways of assessing the quality of an estimator.

Example 2.2.1 (Long Estimation Example)

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \text{Exponential}(\lambda)$ with $n \geq 3$.

We will consider estimating $\tau(\lambda) = 1/\lambda$ and λ.

Example 2.2.1 (cont)

Example 2.2.1 (cont)

Example 2.2.1 (cont)

Example 2.2.1 (cont)

Example 2.2.1 (cont)

Example 2.2.1 (cont)

2.3

Convergence in Probability

Large sample properties

The quality of an estimator often depends on the sample size used to compute the estimator.

We hope that an estimator will tend to be closer to θ as the sample size increases.

To highlight the dependence of an estimator on the sample size, we might include the subscript n in the estimator.

- $\hat{\theta} \equiv \hat{\theta}_n$.
- $\bar{X} \equiv \bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$.

Definition 2.3.1 (Convergence in Probability)

A sequence of random variables $\{X_n\}$ **converges in probability** to a random variable X if, for any $\epsilon > 0$,

$$\lim_{n\to\infty} P(|X_n - X| > \epsilon) = 0.$$

(Equivalently, if $\lim_{n\to\infty} P(|X_n - X| \leq \epsilon) = 1$.)

We write $X_n \xrightarrow{P} X$.

Some Convergence in Probability interpretations

- As n gets large, X_n is almost always close to X.
- As the sample size increases, the probability that X_n is arbitrarily close to X is very high.
- As n gets large, it is very unlikely that X_n differs much from X.

Convergence in probability notes

1. Probabilities are numbers, so the limits are for a sequence of numbers.
2. A constant is a (boring) random variable, so you can have convergence in probability to a number.
3. In practice, we want to know if $\hat{\theta}_n \xrightarrow{P} \theta$.
4. The inequalities can include equals without problems, i.e., $>$ or $\geq$, $<$ or $\leq$.

2.3.1

Markov's Inequality

Generalized Markov Inequality

If X is a random variable and $g(x)$ is a non-negative, real-valued function, then for any $c > 0$,

$$P[g(X) \geq c] \leq \frac{E[g(X)]}{c}.$$

Proof (Generalized Markov Inequality)

Proof (cont)

Proof (cont)

Markov's Inequality

For a random variable X and number $r, c > 0$,

$$P(|X| \geq c) \leq \frac{E(|X|^r)}{c^r}.$$

Proof (Markov’s Inequality)

2.3.2

Chebyshev’s Inequality

Chebyshev's (Tchebychev's) Inequality

Let X be a random variable with mean μ and variance $\sigma^2 < \infty$. For any $k > 0$,

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}.$$

Equivalently,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Proof

2.3.3
The Sample Mean and Convergence in Probability

The Weak Law of Large Numbers (WLLN)

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} F$ with mean μ and variance $\sigma^2 < \infty$. Then

$$\bar{X} \xrightarrow{P} \mu.$$

Proof

Proof (cont)

Proof (cont)

Example 2.3.1

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \text{Exponential}(\lambda)$. What does this tell us about $\bar{X}$?

Example 2.3.2

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \text{Gamma}(\alpha, \beta)$. What does this tell us about $\bar{X}$?

2.3.4

Consistent Estimators

Definition 2.3.2 (Consistent)

Let $\hat{\theta}_n$ be a consistent estimator of θ. Then $\hat{\theta}_n$ is a **consistent estimator** of θ if

$$\hat{\theta}_n \xrightarrow{P} \theta.$$

A question about the consistency of an estimator is related to convergence in probability.

Theorem 2.3.1

An unbiased estimator $\hat{\theta}_n$ of θ is a consistent estimator of θ if

$$\lim_{n \to \infty} \operatorname{var}(\hat{\theta}_n) = 0.$$

Proof

Proof (cont)

Proof (cont)

Definition 2.3.3 (Asymptotically unbiased)

$\hat{\theta}_n$ is an **asymptotically unbiased** estimator of θ if

$$\lim_{n\to\infty} E(\hat{\theta}_n) = \theta.$$

Theorem 2.3.2

An asymptotically unbiased estimator $\hat{\theta}_n$ of θ is a consistent estimator of θ if

$$\lim_{n \to \infty} \operatorname{var}(\hat{\theta}_n) = 0$$

Proof

Proof (cont)

Proof (cont)

Theorem

An estimator $\hat{\theta}_n$ of θ is a consistent estimator of θ if

$$\lim_{n \to \infty} \mathrm{MSE}(\hat{\theta}_n) = 0$$

Proof

Example 2.3.3

Let $X_1, X_2, \dots, X_n \overset{i.i.d.}{\sim} \text{Uniform}(0, \theta)$. Show that $X_{(n)} \xrightarrow{P} \theta$.

Example 2.3.3 (cont)

Example 2.3.3 (cont)

Example 2.3.3 (cont)

Example 2.3.3 (cont)

Example 2.3.3 (cont)

2.3.5

Things About Convergence in Probability That Will Not Surprise You

Theorem 2.3.3

Suppose that $\{X_n\}$ and $\{Y_n\}$ are sequences of random variables such that $X_n \xrightarrow{P} X$ and $Y_n \xrightarrow{P} Y$ for some random variables X and Y. Then

1. $X_n + Y_n \xrightarrow{P} X + Y$.
2. $X_n Y_n \xrightarrow{P} XY$.

Theorem 2.3.3 (cont)

3. $X_n/Y_n \xrightarrow{P} X/Y$ as long as the denominators are non-zero with probability 1.
4. Continuous Mapping Theorem: For any continuous function g,

$$g(X_n) \xrightarrow{P} g(X).$$

Proof

Proof (cont)

Proof (cont)

Proof (cont)

Proof (cont)

Proof (cont)

Example 2.3.4

Suppose the $X_n \xrightarrow{P} b$ and a is a constant. Does aX_n converge?

Example 2.3.4 (cont)

Example 2.3.5

Let $\{a_n\}$ be a sequence of real number such that

$$\lim_{n\to\infty} a_n = a.$$

Show that $a_n \xrightarrow{P} a$.

Example 2.3.5 (cont)

Example 2.3.6

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \text{Uniform}(0, \theta)$. Use Theorem 2.3.1 to show that $X_{(n)} \xrightarrow{P} \theta$.

Example 2.3.6 (cont)

Example 2.3.6 (cont)

2.4

Convergence in Distribution

Definition 2.4.1 (Convergence in Distribution)

Let $\{X_n\}$ be a sequence of random variables with cdfs $F_n(x) = P(X_n \leq X)$. Let X be a random variable with cdf $F(X)$. X_n **converges in distribution** to X if

$$\lim_{n \to \infty} F_n(x) = F(x),$$

for all X where F is continuous.

We denote this property as $X_n \xrightarrow{d} X$.

Convergence in Distribution

The convergence in distribution of X_n to X may also be denoted as $X_n \xrightarrow{D} X$.

Convergence in distribution is sometimes called **convergence in law** and denoted $X_n \xrightarrow{L} X$.

Example 2.4.1

Let $X_1, X_2, \dots, X_n \overset{i.i.d.}{\sim} \text{Pareto}(1)$. Define $Y_n = nX_{(1)}$. The cdf of Y_n converges to what distribution?

Example 2.4.1 (cont)

Example 2.4.1 (cont)

Example 2.4.1 (cont)

Example 2.4.2

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \text{Uniform}(0, 1)$. Consider $Y_n = X_{(n)}$. The cdf of Y_n converges to what?

Example 2.4.2 (cont)

Example 2.4.2 (cont)

Example 2.4.2 (cont)

Notes

CDFs are always right-continuous functions.

Convergence in distribution only must be done where $F(x)$ is continuous.

2.4.1

Convergence in Probability is Stronger

Theorem 2.4.1

$$X_n \xrightarrow{P} X \Rightarrow X_n \xrightarrow{d} X.$$

Proof

Proof (cont)

Proof (cont)

Example 2.4.4

$$X_n \xrightarrow{P} X \not\Rightarrow X_n \xrightarrow{d} X.$$

Example 2.4.4 (cont)

Example 2.4.4 (cont)

Example 2.4.4 (cont)

Theorem 2.4.2

Suppose that $X_n \xrightarrow{d} c$ where $c \in \mathbb{R}$. Then $X_n \xrightarrow{P} c$.

Proof

Proof (cont)

Proof (cont)

2.4.2

The Continuous Mapping Theorem

The Continuous Mapping Theorem

Suppose that $X_n \xrightarrow{d} X$ and g is a continuous function. Then

$$g(X_n) \xrightarrow{d} g(X).$$

The proof is omitted because it is long.

Example 2.4.4

Let $X, X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$. Then $X_n \xrightarrow{d} X$.

This is why convergence in distribution is weak.

Example 2.4.4 (cont)

Example 2.4.4 (cont)

Unexpected property

Even if $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$, it is **not necessarily true** that $X_n + Y_n \xrightarrow{d} X + Y$.

Example 2.4.5

Suppose that $X, X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} N(0,1)$ and $Y_n := -X_n$. To what does $X_n + Y_n$ converge in distribution?

Example 2.4.5 (cont)

Example 2.4.5 (cont)

Example 2.4.5 (cont)

2.4.3

Slutsky's Theorem: Mixing Convergence Types

Slutsky's Theorem

Suppose that $X_n \xrightarrow{P} a$, where $a \in \mathbb{R}$ and $Y_n \xrightarrow{d} Y$. Then

1. $X_n + Y_n \xrightarrow{d} a + Y$.
2. $X_n Y_n \xrightarrow{d} aY$.
3. $Y_n / X_n \xrightarrow{d} Y/a$ if $a \neq 0$.

Proof

Proof (cont)

Proof (cont)

Proof (cont)

Proof (cont)

Example 2.4.6

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \text{Pareto}(1)$. We know from Example 2.4.1 that $Y_n := nX_{(1)} \xrightarrow{d} Y$ where $Y \sim \text{Exponential}(1)$. To what does $X_{(1)}$ converge in distribution?

Example 2.4.6 (cont)

Example 2.4.6 (cont)

2.4.4

Convergence in Distribution and Moment Generating Functions

Theorem 2.4.3

Let $X_1, X_2, \ldots, X_n$ be a sequence of random variables. Let $F_n(x)$ be the cdf of X_n and $M_n(t)$ be the mgf of X_n. Let X be a random variable with cdf $F(x)$ and mgf $M(t)$.

Then

$$\lim_{n\to\infty} M_n(t) = M(t) \Rightarrow \lim_{n\to\infty} F_n(x) = F(x).$$

i.e., if $\lim_{n\to\infty} M_n(t) = M(t)$ then $X_n \xrightarrow{d} X$.

Notes

1. The first limit for the mgf in Theorem 2.4.3 only needs to hold in some open interval containing zero.
2. The result would be if and only if (go both directions) if the mgf always existed (it might diverge to infinity).

Example 2.4.7

Let $X_1, X_2, \ldots$ be a sequence of independent random variables with

$$X_n \sim \text{Binomial}(n, \lambda/n).$$

To what does X_n converge in distribution?

Example 2.4.7 (cont)

Example 2.4.7 (cont)

Example 2.4.7 (cont)

Example 2.4.7 (cont)

2.5

The Central Limit Theorem

The Central Limit Theorem (CLT)

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} F$ with mean μ and variance $\sigma^2 < \infty$. Define

$$Z_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

Then as $n \to \infty$, $Z_n \xrightarrow{d} Z$ where $Z \sim N(0, 1)$.

Notes about the CLT

1. The CLT requires a finite variance.
2. Our proof will assume the mgf exists, which isn't always satisfied by distributions with finite variance.
3. The proof can be completed more generally using characteristic functions, which always exist (and are an extension of mgfs involving imaginary numbers).

2.5.1

Proof of the Central Limit Theorem

Proof

Proof (cont)

Proof (cont)

Proof (cont)

2.5.2

Asymptotic Normality

Definition 2.5.1 (Asymptotically Normal)

X_n has an **asymptotically normal** distribution with mean μ_n and variance σ_n^2 if

$$\frac{X_n - \mu_n}{\sigma_n} \xrightarrow{d} N(0,1).$$

Notation

If X_n has an **asymptotically normal** distribution with mean μ_n and variance σ_n^2, we will write

$$X_n \overset{asymp}{\sim} N(\mu_n, \sigma_n^2).$$

Note: It is not correct to write $X_n \xrightarrow{d} N(\mu_n, \sigma_n^2)$ since the first part implies that $n \to \infty$, so there shouldn't be any ns on the right side of the arrow.

Example 2.4.1

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} F$ with mean μ and variance $\sigma^2 < \infty$. What can we say about the asymptotic distribution of $\bar{X}$?

Example 2.4.2

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} F$ with mean μ and variance $\sigma^2 < \infty$. What can we say about the asymptotic distribution of $\sum_{i=1}^{n} X_i$?

2.5.3

Computing Probabilities for the Normal Distribution

Integration difficulties

If $X \sim N(\mu, \sigma^2)$, then

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right).$$

Proving that $f(x; \mu, \sigma^2)$ integrates to 1 requires the use of polar coordinates.

Computing $F(x; \mu, \sigma^2) = \int_{-\infty}^{x} f_X(s; \mu, \sigma^2)\, ds$ requires numerical evaluation.

Z-tables

Let $Z \sim N(0,1)$.

The cdf of Z is defined as

$$F(z) = P(Z \leq z) = \Phi(z).$$

$\Phi(z)$ can be evaluated easily in calculators or statistical software.

E.g., here is how we evaluate $\Phi(1)$ in R:

```
pnorm(1)
```

```
[1] 0.8413447
```

Relating Z to $N(\mu, \sigma^2)$

If $X \sim N(\mu, \sigma^2)$, then

$$Z := \frac{X - \mu}{\sigma} \sim N(0, 1).$$

Similarly, if $Z \sim N(0, 1)$, then

$$X := \mu + \sigma Z \sim N(\mu, \sigma^2).$$

Computing a probability

Let $X \sim N(1, 2)$.

$$
\begin{aligned}
P(X \leq 2.3) &= P\left(\frac{X-1}{\sqrt{2}} \leq \frac{2.3-1}{\sqrt{2}}\right) \\
&= P(Z \leq 0.92) \\
&= \Phi(0.9192) \\
&= 0.8210.
\end{aligned}
$$

```r
pnorm((2.3-1)/sqrt(2))
```

```
[1] 0.8210147
```

2.5.4

A Numerical Example for the CLT

CLT numerical example

Let $\bar{X}$ denote the sample mean of a random sample of size 100 from the $\chi^2(50)$ distribution.

Estimate $P(49 < \bar{X} < 51)$.

CLT numerical example (cont)

CLT numerical example (cont)

2.5.5

The Delta Method

Delta Method introduction

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} F$ with mean μ and variance $\sigma^2 < \infty$.

By the WLLN, we know that $\bar{X} \xrightarrow{P} \mu$.

That implies that $\bar{X} \xrightarrow{d} \mu$

By the Continuous Mapping Theorem (CMT), we can say that $\bar{X}^2 \xrightarrow{d} \mu^2$

What can we say about the asymptotic distribution of $\bar{X}^2$?

More information

The CLT says that

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

Thus, $\bar{X} \overset{asymp}{\sim} N(\mu, \sigma^2/n)$.

Can we say $\bar{X}^2 \overset{asymp}{\sim} N(a_n, b_n)$ for some $\{a_n\}$ and $\{b_n\}$?

The Delta Method

Suppose that we have a sequence of estimators $\hat{\theta}_n$ for θ such that

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma^2).$$

If $g : \mathbb{R} \to \mathbb{R}$ is continuously differentiable in a neighborhood of θ, then

$$\sqrt{n}(g(\hat{\theta}_n) - g(\theta)) \xrightarrow{d} N(0, \sigma^2[g^{'}(\theta)]^2).$$

Proof

Proof (cont)

Proof (cont)

Proof (cont)

Example 2.5.3

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} F$ with mean μ and variance $\sigma^2 < \infty$. Determine the asymptotic distribution of $\bar{X}^2$.

Example 2.5.3 (cont)

Example 2.5.3 (cont)

2.6

The Sample Variance

The Sample Variance

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} F$ with mean μ and variance $\sigma^2 < \infty$.

The **sample variance** of $\{X_1, X_2, \ldots, X_n\}$ is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

Facts about the sample variance

- $E(S^2) = \sigma^2$.
- $\mathrm{var}(S^2) = \frac{1}{n} E[(X-\mu)^4] - \frac{(\sigma^2)^2(n-3)}{n(n-1)}$.
- If $E[(X-\mu)^4] < \infty$, then $S^2 \xrightarrow{P} \sigma^2$.